

Sanchit Chaudhary

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SUMMARY

Data professional with over 4 years of experience in data science and analytics, holding an MS in Data Science. I am skilled in designing and optimizing ETL pipelines, performing data analytics, and implementing machine learning solutions. Demonstrates expertise in Python, SQL, Spark, Gen-AI, Dbt and data modelling, along with strong proficiency in cloud platforms (AWS). A competitive individual with a background in state-level wrestling, debate, and badminton showcasing discipline, critical thinking, teamwork.

EDUCATION

University of South Florida, Tampa, FL

MS in Data Science

Graduation: May 2025

GPA: 3.93/4.0

Kurukshetra University, India

B.Tech in Computer Science

August 2016 – May 2020

GPA: 3.90/4.0

Relevant Coursework: Machine Learning, Database Management, Big Data, Data Warehousing, Cloud Computing

EXPERIENCE

Data Engineer Research Assistant | University of South Florida, Tampa

September 2024 – Present

- Designed AWS EC2 Remote Desktop with Docker, Spark and Hive, providing 500+ students centralized access to datasets and analytics tools (SQL, Python, Tableau).
- Educated and mentored over 120 students in advanced statistics concepts (T-tests and Chi-squared tests, regression analysis, A/B testing) through weekly labs, fostering a collaborative learning environment improving class performance by 35%.
- Developed Power BI dashboards for automated data pipelines, reducing report generation time by 30% and empowering 10+ academic teams to monitor student academic performance.

Data Science Intern | Amgen, California

May 2024 – August 2024

- Developed a Central Orchestrator (AI agent) integrated into MS Teams channel, significantly improving query resolution time from 1 week time to mere minutes, thus enhancing operational efficiency for 3500 downstream users.
- Designed a Power BI dashboard (DAX, time series forecasting) to visualize and predict user query demand trends, enabling proactive capacity planning for an AI chatbot, also built the system design document for the AI agent.
- Utilized Gen-AI tools including AI Studio's with OpenAI LLM solutions to implement a Retrieve Augment Generation (RAG) framework, enhancing LLM accuracy by 25% for queries based on org's healthcare data with advanced prompt engineering.
- Integrated Power Automate flows with backend Databricks code, achieving an 80% deflection rate in user queries and projecting annual savings of \$110K by automating routine tasks.

Data Engineer II | Tata Consultancy Services | Client: Bank of America

July 2021 – June 2023

- Developed ETL data pipelines for various downstream analytics applications like ML modelling, data analysis, and BI report generation, supporting a \$10M intelligent voice and chat-driven financial concierge application called "Erica".
- Utilized Hadoop tech stack like Apache Sqoop, Flume to extract data from various sources (databases, APIs) in different formats (CSV, Avro, Parquet), Spark to transform, and External Hive tables as warehouse.
- Optimized SQL queries using CTEs, joins, grouping, aggregation, and window functions for complex data extraction.
- Automated batch ETL pipeline using Apache Oozie to extract new financial data from relational Database and applied transformation like Denormalization, column name change, null values handling and loaded to HDFS for PowerBI reporting.
- Worked on Data mapping, logical data modeling, and created class diagrams and ER diagrams.
- Worked on POC for migrating on-prem data pipelines to cloud (AWS), lowering infrastructure costs by 22%.

Data Engineer | Tata Consultancy Services | Client: Bank of America

January 2020 – June 2021

- Conducted thorough reviews and analysis of business requirements, collaborating with cross-functional teams, translating insights into actionable data that contributed to a \$1.2 million increase in revenue.
- Led migration of data workflows from SSMS to Snowflake using dbt, improving processing efficiency by 35%.
- Built and enhanced dbt models to streamline data transformation, decreasing errors by 25%.
- Utilized SQL, Python, and advanced Excel for data cleaning and analysis to ensure high-quality datasets available, resulting in a 20% reduction in data-related errors.
- Created dynamic Tableau dashboards with calculated fields and LOD expressions to track essential banking KPIs including (Spendpath, Low balance Watch, Bill Reminders etc.) enabling stakeholders to drill down into critical metrics effectively.

PROJECTS

Advanced Database Management - Final Project | PostgreSQL, Python, Airflow, Power BI

October 2023

- Designed an ETL pipeline with PostgreSQL, Python and Airflow, using Uber Fares Dataset from Kaggle.
- Built a Tableau dashboard for data analysis, showcasing trends in ride fares, distance and times across multiple cities.

Neural Image Clustering System (NICS) | Python, PyTorch, ResNet50

March 2022

- Created an image clustering system, using PyTorch, capable of clustering and filtering images based on the objects present.
- Deployed an ensemble model, combining their output tensors to assign images to clusters based on occurrence probabilities.

Other Projects: Realtime Data Streaming (Kafka), Power of Woman (Tableau), Spam Classifier (NLP) (all on GitHub)

TECHNICAL SKILLS

Languages: Python, R, SQL, C++, Shell Script

AI & Data Science: Scikit-Learn, PyTorch, TensorFlow, Matplotlib, Seaborn, NumPy, Pandas, SciPy, Snowflake

Data Engineering: AWS, GCP, Apache Kafka, Snowflake, Databricks, Airflow, PySpark, BigQuery, ETL Processes, Big Data

Databases: SQL, PostgreSQL, MongoDB, Hive, Data Warehousing, Data Modeling, AWS Redshift, GCP, DBT

Business Intelligence: Tableau, Power BI, Looker, Google Data Studio, Qlik, Statistics, A/B Testing, MS Excel

Development Tools: Docker, Kubernetes, Jenkins, VMs, GULP, CLI, Git, Bitbucket, MATLAB